Elliptical Target Measurements by Matching: Implementing and Testing, Least Squares Matching of DF Imagery for the Precise Estimation of Distortion Parameters

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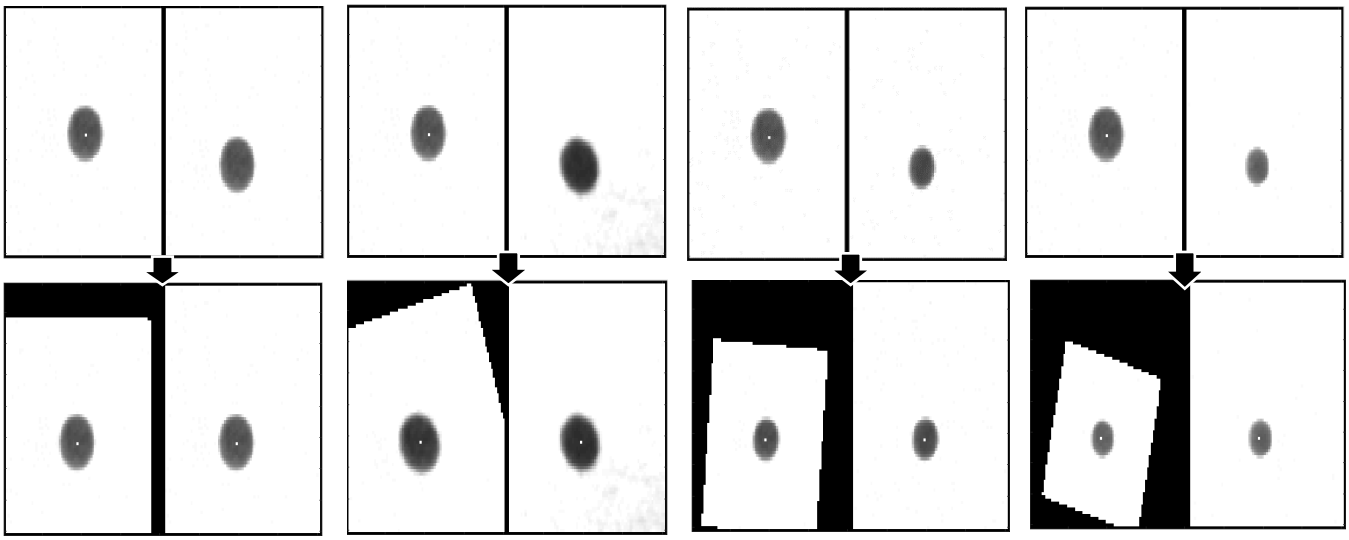


Figure 1. A collection of matched elliptical targets. The left target moves, the right target does not. The matching starts at the upper images and finishes at the lower images. The left target (target 5) on each sub-image, was matched to the right target. The center of target 5 was known, so the center of the right images were found by estimating geometric distortions between left and right images.

Preamble

Accurate reconstruction of bone translations and rotations from dual-fluoroscopic (DF) data requires accurate calibration of the system imaging geometry and the many imaging distortions that corrupt the data. Recent research has demonstrated that the procedure is best performed with a three-dimensional calibration object comprising radio-opaque spherical targets. Success of the calibration requires strong imaging geometric design and a precise and robust algorithm for measurement of the target centres in the imagery. The current method for target measurement (ellipse-fitting) is sub-optimal, yielding coordinate precision only at the level of 0.3 to 0.4 pixels. This project focused on improving the target measurement precision to better than 0.1 pixels with least-squares template matching (Lichti, 2015). The report sums up the entirety of the project by presenting:

- A methodology to re-implement least-squares matching.
- The feasibility of least squares matching in an automatic calibration system.

A. Least-Squares Matching Defined

Least-squares matching was an image matching method used to estimate the distortion parameters between two correlated images. It was introduced by and implemented before in (Gruen, 1985). Any automated system for three-dimensional point positioning must include a potent procedure for stereo image or dual image matching (Gruen, 1996). The paper was used to develop a concise and easy to implement algorithm to automate the DF stereoscopic calibration setup. Also, the reliability, feasibility, and rigor of least squares matching for the setup was raised.

Least-squares matching can be applied to grey images, which made it ideal for DF imagery. Least-squares follows the fundamentals of:

- Analyzing a situation.
- Deriving a functional model.
- Understanding the form of the functional model.
- Applying the least-squares estimation.
- Obtaining the statistics and visuals.

These fundamentals were applied to estimate the distortions between sample images acquired from the DF imagery. These samples include:

(a) Single target sub-images drawn from different conditions in terms of target shape and radiometric quality.

(b) Single target template images that were artificially generated.

(c) Multiple target sub-images from samples of large and small target sizes.

(d) Sequential knee images taken during locomotion.

The images were either sequential or similar; which implied high signal correlation. Given two images, one image was the image that would be adjusted and matched to the other. This image was called **measured** (adjusted image). The other unchanging or reference image was called **true**. Further, the grey value differences between measured and true described the geometric and radiometric parameters. The geometric distortion parameters described motion that transforms the measured image plane to the true image plane. The radiometric anomaly parameters described radiant energy that transform the measured pixel grey values to the true pixel grey values. Therefore, the differences between the left and right targets of the top half of Figure 1 were functionally modelled by the distortion parameters.

B. Target Measurements and Distortion Parameters

Any image point on the measured image may be tracked as it is geometrical transformed to the true image. For elliptical target images, the ellipse center was tracked as the first image matched to the next. Given a sequence of elliptical target images, the ellipse center on the first image can be tracked to the next images. This capability was necessary for automation of a calibration setup.

There will be high signal and spectrum correlation between two sequential images from a Dual Fluoroscopy video. This correlation was the basis for least-squares matching. Also, it was important to note that sequential images given by a Dual Fluoroscopy system were subject to a greater amount of radiometric and geometric distortions compared to standard camera images. The geometric distortions were caused by going from 3D to 2D, the shape of the object, relative positioning of the camera and object, perspective imaging and optical lens defects. Electromagnetic radiation radiating off the object to create the image was affected by the dual fluoroscopic system; specifically, transmission media and the receiver.

Using a functional model, we can define an image and its proportionality and correlation with the next image. We get a set of geometric distortion parameters and radiometric anomaly parameters to match the sequential images. Using least squares estimation, one may calculate these parameters and match pixel points from one image to the next. This matching is not limited to just sequential images. The least-squares match was successful with enough signal and spectrum correlation between a template image and another sub-image. The program can match features on a 2D representation of a 3D object template, to the same features on the 2D image of the 3D object, while estimating the geometric and radiometric distortion parameters. Being able to match images is the same as being able to track features between images. So, this procedure can also track how targets in the calibration are moving. The computational workflow implemented in MATLAB can be seen in figure 2.

METHODOLOGY

Initialize

Known image as $f = f(u, v)$
Unknown image as $g = g(u, v)$
Parameters as $x = [a_0\ a_1\ a_2\ b_0\ b_1\ b_2\ r_0\ r_1]^T$
Initial $x(1:6)$ as $x_g^o = [0\ 1\ 0\ 0\ 0\ 1]^T$
Set thresholds on dx_i as t_i

Non-linear Parametric Least Squares for Geometric Distortion Parameters $x_g = x(1:6)$

$[g_x, g_y] \leftarrow$ ImGradient(g)
$A_g \leftarrow$ ReturnA(g, g_x, g_y)
loop:
  $(x, y) \leftarrow$ AffineTrans($(u, v), x_g^o$)
  $f_{adj}(x, y) \leftarrow$ Interpolant(f, (x, y))
  $f_{adj}(u, v) \leftarrow$ Interpolate($f_{adj}(x, y), (u, v)$)
  $im_{bool} \leftarrow$ **not** (isnan($f_{adj}(u, v)$))
  $dx_g \leftarrow$ ParaLS(A_g, g, $f_{adj}(u, v)$) @im_{bool}
  $x_g^o \leftarrow x_g^o + dx_g$
  if $t_i > |dx|$ **or** iters > max
    break
$g(u, v) \leftarrow$ Interpolant(g, (u, v))
$g(x, y) \leftarrow$ Intepolate($g(u, v), (x, y)$)
$im_{bool} \leftarrow$ not (isnan($g(x, y)$))
$e_g \leftarrow f(u, v) - g(x, y)$ @im_{bool}
ErrorAnalysis(e_g, f, g, $g(x, y)$)

Linear Parametric Least Squares for Radiometric Anomaly Parameters $x_r = x(7:8)$

$A_r \leftarrow$ ReturnA($g(x, y)$) @im_{bool}
$w \leftarrow -f$ @im_{bool}
$x_r \leftarrow (A_r^T A_r)^{-1} A_r^T w$
$g_{adj}(x, y) \leftarrow$ RadioTrans($g(x, y), x_r$)
$e_r \leftarrow f - g_{adj}(x, y)$
ErrorAnalysis(e_r, f, g, $g_{adj}(x, y)$)

Output

$x = [x_g, x_r]$
Given a known point on f, the same point on g is estimated by x_g. For example, (x_c, y_c) is the center point on f, then center point on g is given by
$(x_c', y_c') =$ AffineTrans($(x_c, y_c), x_g$)

Figure 2. Computational workflow of least squares matching.

C. DF Imagery as Raw Data

Grayscale DF images were the raw data; and, it is presented in figure 3. Sub-images were taken as patches isolating target(s). Grayscale images are described by $h(u, v) = image$. Where, (u, v) are pixel locations and h is the corresponding brightness values.

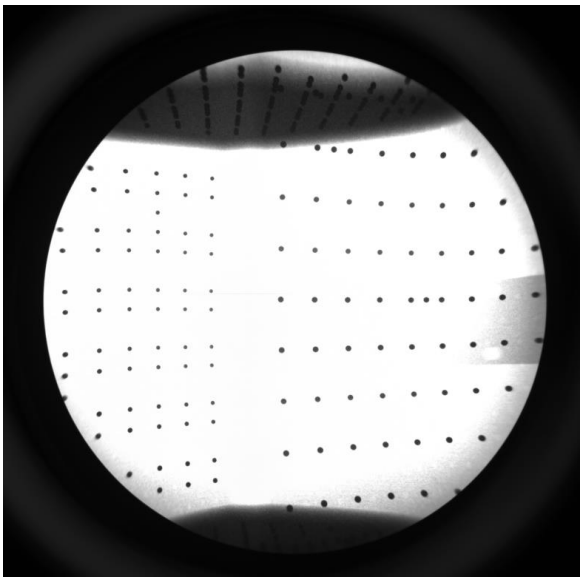


Figure 3. Sample cubic calibration full image. Refer to Appendix I.

D. The Situation

The situation displayed in Figure 4 is that the left image must match to the right image. The left image is a target image which closely resembles a circle. The right image is distorted and clearly deviates from a circle. So, a least squares methodology is used to estimate the geometric distortions and radiometric anomalies between them. The geometric distortions are described by the corresponding image areas having different coordinates/distributions in their image planes. The radiometric anomalies are described by collinear image pixels having different brightness values. Due to the nature of the problem and available mathematical models, the radiometric adjustment is completed after the geometric adjustment successfully matches the images.

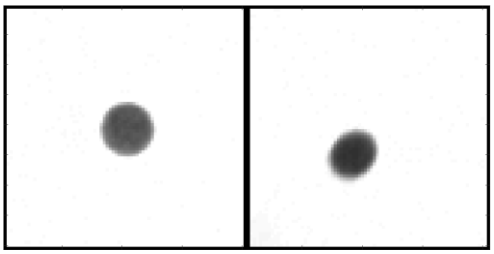


Figure 4. Left target image (Target 5) is taken as measured and is described by the image function f(u,v). The right target image is the taken as true and is described by the image function g(u,v). Measured matches to true.

$$\bar{l}_{meas} = f(u, v) \quad (1)$$

$$\bar{l}_{true} = g(u, v) \quad (2)$$

The true measurements are simply what measurement would be if they were error free. The true images are taken as *(a) single target sub-images, (c) the next multiple target sub-image in sequence*, and *(d) the next knee image in sequence*. The true measurements are not adjusted, so they describe the situation **consistently**. Further, setting them as the visually more distorted image means that they would not have to be transformed. Again, the right image is $\bar{l}_{true}$ and they consistently describe the relationship between distortions parameters $\bar{x}$ and measurements.

Measurements have random and systematic errors with respect to the true. The systematic errors are the geometric and radiometric distortions which are functionally modelled and estimated. The measured images are taken as Target 5, generated template images, the first multiple target sub-image in sequence, and the first knee image in sequence. The random errors are the noise which come from image capturing and become minimized. Again, the left image is $\bar{l}_{meas}$ and always contains random error given by $\bar{e}$. Generated

$\#\ measurements,$ n $= \#pixels\ of\ l_{meas}$

$\#\ equations,$ r $= \#pixels\ of\ l_{adj}$ * r varies during adjustment

$\#\ unknowns,$ $u_g = \#geometric\ parameters$

$\#\ unknowns,$ $u_r = \#radiometric\ parameters$

From here on out, the subscripts $_g$, and $_r$ refer to the geometric and radiometric entities, respectively.

E. The Geometric Functional Model

Given coordinates (u, v) as image pixel locations within m columns and n rows

$$\{u \mid 1 \leq u \leq m, u \in \mathbb{N}\}$$

$$\{v \mid 1 \leq v \leq n, v \in \mathbb{N}\}$$

and, the coordinates (x, y) as any point in the image plane

$$\{x \mid 1 \leq x \leq m, x \in \mathbb{R}\}$$

$$\{y \mid 1 \leq y \leq n, y \in \mathbb{R}\}$$

then,

"f at (u, v) equals g at some (x, y) plus noise"

For known pixel locations (u, v), transformation parameters that relate (u, v) to the unknown (x, y) are estimated. These concepts are the basis for deriving and iterating the geometric functional model. The basis is equated as follows

$$f(u, v) - \bar{e} = g(x, y) \quad (3)$$

$g(x, y)$ must be linearized because $\bar{x}_g$ is implicitly described by the non-linear function g. Plotting g clearly shows its non-linearity.

$$f(u, v) - \bar{e} = g(x^o, y^o) + \frac{\partial g(x^o, y^o)}{\partial x} dx + \frac{\partial g(x^o, y^o)}{\partial y} dy \quad (4)$$

The geometric distortions between images are described by an affine movement. In fact, the adjustment applied to l_{meas} is an affine image transformation (Gruen, 1985). Therefore, the geometric math model is given by

$$x = a_0 + a_1 u + a_2 v \quad (5)$$

$$y = b_0 + b_1 u + b_2 v \quad (6)$$

where x and y are initial pixel locations, u and v are transformed image locations, a_0 and b_0 translate images, a_1 and b_2 scale images along the x-pixel and y-pixel axis, respectively, and a_2 and b_1 shear images. Further, some combinations of a_1, b_2, a_2 and b_1 rotate images.

The full set of least squares observables are

$$\bar{x}_g = [a_0\ a_1\ a_2\ b_0\ b_1\ b_2]^T$$

This math model should only be used for the corresponding planar geometric distortions. Given a fitting situation, any corresponding planar math model may be used. Further, the feasibility of an affine transformation can be determined.

When the model assumed is overparameterized (too complex relative to the true underlying model), unnecessary sampling variance is introduced from estimation of extra parameters. This added variance may increase correlation and compromise the accuracy of the match (Gruen, 1996).

When the model assumed is under parametrized, the full phenomena of the design is insufficiently modelled, which can create a bias in the error; meaning, there is not a random normal distribution. Further, this bias may result in an increase in the precision statistics of estimated parameters; this will be unreliable (Gruen, 1996).

Images that need to be rotated often have higher correlations of a_1, b_2, a_2 and b_1. Given circular and elliptical target images, this correlation was hypothesized to be even higher because 6 affine transformation would over-parametrize certain matches. Therefore, it was necessary to test

combinations of affine transformation parameters. Tested combinations include:

- Translations, $\bar{x}_g = [a_0\ b_0]^T$
- Translations and scales, $\bar{x}_g = [a_0\ a_1\ b_0\ b_2]^T$
- Translations, scales and rotation:
  - $\bar{x}_g = [a_0\ a_1\ b_0\ b_1\ b_2]^T$
  - $\bar{x}_g = [a_0\ a_1\ a_2\ b_0\ b_2]^T$
- Full affine transformation, $\bar{x}_g = [a_0\ a_1\ a_2\ b_0\ b_1\ b_2]^T$

When two images, such as images in Figure 2, are matched, the least-squares has no problem seeing that the target on the left image must match to the only available target in the right image; there is no ambiguity. So, initial approximations for the observables can be

$$\bar{x}_g^o = [0\ 1\ 0\ 0\ 0\ 1]^T$$

However, when matching images which have ambiguous features. The initial a priori parameters should be set properly. This allows the program to know which ellipse matches to which. For figure 3, the initial parameters should be set as

$$\bar{x}_g^o = [-30\ 1\ 0\ 0\ 0\ 1]^T$$

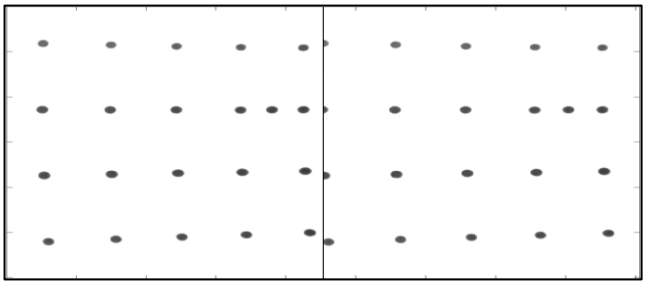


Figure 5. Matching of mutiple target sub-images. It is unclear which ellipse in left image must fit which ellipse in the right image.

Although clear, total differentials are with respect to the unknown parameters, and not u and v because u and v are with are apart of the deterministic, consistent part of the least-squares adjustment. So, total differentials to find dx and dy are

$$dr = \sum \frac{\partial r}{\partial p_i} dp_i$$

$$dx = da_0 + uda_1 + vda_2 + \cancel{a_1 du} + \cancel{a_2 dv}$$

$$dy = db_0 + udb_1 + vdb_2 + \cancel{b_1 du} + \cancel{b_2 dv}$$

The A matrix for the full set of six geometric parameters is given by

$$A_{g_{rx6}} = \begin{bmatrix} g_x(x_1^o,y_1^o) & 1g_x(x_1^o,y_1^o) & 1g_x(x_1^o,y_1^o) & g_y(x_1^o,y_1^o) & 1g_y(x_1^o,y_1^o) & 1g_y(x_1^o,y_1^o) \\ g_x(x_1^o,y_2^o) & 1g_x(x_1^o,y_2^o) & 2g_x(x_1^o,y_2^o) & g_y(x_1^o,y_2^o) & 1g_y(x_1^o,y_2^o) & 2g_y(x_1^o,y_2^o) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ g_x(x_1^o,y_n^o) & 1g_x(x_1^o,y_n^o) & ng_x(x_1^o,y_n^o) & g_y(x_1^o,y_n^o) & 1g_y(x_1^o,y_n^o) & ng_y(x_1^o,y_n^o) \\ g_x(x_2^o,y_1^o) & 2g_x(x_2^o,y_1^o) & 1g_x(x_2^o,y_1^o) & g_y(x_2^o,y_1^o) & 2g_y(x_2^o,y_1^o) & 1g_y(x_2^o,y_1^o) \\ g_x(x_2^o,y_2^o) & 2g_x(x_2^o,y_2^o) & 2g_x(x_2^o,y_2^o) & g_y(x_2^o,y_2^o) & 2g_y(x_2^o,y_2^o) & 2g_y(x_2^o,y_2^o) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ g_x(x_m^o,y_n^o) & mg_x(x_m^o,y_n^o) & ng_x(x_m^o,y_n^o) & g_y(x_m^o,y_n^o) & mg_y(x_m^o,y_n^o) & ng_y(x_m^o,y_n^o) \end{bmatrix}$$

where n is the number of image rows, m is the number of image columns, $p = n * m$, and g_x and g_y are partial derivatives. Given that a geometric parameter is not used, the corresponding column should be removed.

Partial derivatives g_x and g_y were derived using an image window at each pixel location (x^o, y^o) of g. The partial derivative is described as the change in brightness values over the change in pixel values. Figure 6 was used to derive directional gradients at each (x^o, y^o).

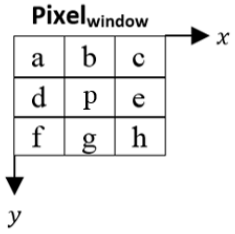


Figure 6. The pixel window used to calculate partial derivatives at p = (x°,y°).

$$g_x = \frac{\partial g(x^o,y^o)}{\partial x} = \frac{\left(\frac{g(\mathrm{e})}{d_{\mathrm{pe}}} + \frac{g(\mathrm{c})}{d_{\mathrm{pc}}} + \frac{g(\mathrm{h})}{d_{\mathrm{pd}}}\right) - \left(\frac{g(\mathrm{d})}{d_{\mathrm{pd}}} + \frac{g(\mathrm{a})}{d_{\mathrm{pa}}} + \frac{g(\mathrm{f})}{d_{\mathrm{pf}}}\right)}{d_{\mathrm{de}}} \quad (7)$$

$$g_y = \frac{\partial g(x^o,y^o)}{\partial y} = \frac{\left(\frac{g(\mathrm{g})}{d_{\mathrm{pg}}} + \frac{g(\mathrm{h})}{d_{\mathrm{ph}}} + \frac{g(\mathrm{f})}{d_{\mathrm{pf}}}\right) - \left(\frac{g(\mathrm{b})}{d_{\mathrm{pb}}} + \frac{g(\mathrm{c})}{d_{\mathrm{pc}}} + \frac{g(\mathrm{a})}{d_{\mathrm{pa}}}\right)}{d_{\mathrm{bg}}} \quad (8)$$

$$d_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2} \quad (9)$$

Due to the geometry of an image, directional gradients cannot be calculated at edge pixels. So, g_x and g_y are images that have 2 less rows and columns than $g(x, y)$.

The nature of matching images allowed for various geometric adjustment methods. To repeat, each method follows the theory that

"f at (u, v) equals g at some (x, y) plus noise"

However, method 2 was used to obtain the results.

METHOD 1 Conventional Parametric Least-Squares

This method follows convectional least squares principles. Computationally, it is the most difficult to implement. Affine transformations are applied to (x^o, y^o) and the A_g matrix is updated iteratively. Neither, f or g are adjusted. However, at each iteration they both need to be interpolated. So, images

are never adjusted, but their grey value pixel locations (u, v) are interpolated to some (x, y). The geometric parameters are adjusted at each iteration. The full non-linear parametric least-squares is as follows:

1: $f(u,v) - \bar{e} = g(x,y)$

2: $f(u,v) - \bar{e} = g(x^o, y^o) + \frac{\partial g(x^o,y^o)}{\partial x} dx + \frac{\partial g(x^o,y^o)}{\partial y} dy$
- $\begin{bmatrix} x^o \\ y^o \end{bmatrix} = \begin{bmatrix} a_1^o & a_2^o \\ b_1^o & b_2^o \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} a_0^o \\ b_0^o \end{bmatrix}$
- $dx = da_0 + uda_1 + vda_2$
- $dy = db_0 + udb_1 + vdb_2$
- g_x from (4) for the first iteration
- g_y from (5) for the first iteration
- f must be linearly interpolated to pixel locations (x^o, y^o), which is just (u, v) for the first iteration.

3: $f(x^o,y^o) - \bar{e} = g(x^o,y^o) + g_x da_0 + g_x \bar{u} da_1 + g_x \bar{v} da_2 + g_y db_0 + g_y \bar{u} db_1 + g_y \bar{v} db_2$

4: $-\bar{e} = \bar{w} + A_g \bar{\delta}$
- $\bar{w} = g(x^o,y^o) - f(x^o,y^o)$
- $\bar{\delta} = [da_0\ da_1\ da_2\ db_0\ db_1\ db_2]^T$
- $A_g = A_g$ from (5)

5: $\bar{e}^T \bar{e} = \left(\bar{w} + A_g \bar{\delta}\right)^T (\bar{w} + A_g \bar{\delta}) = min$

6: $\bar{\delta} = -\left(A_g^T A_g\right)^{-1} (A_g^T \bar{w})$

7: $\bar{x}_g = \bar{x}_g^o + \bar{\delta}$

8: $\bar{x}_g^o = \bar{x}_g$

9: Update terms
- $\bar{x}_g = \bar{x}_g^o + \bar{\delta}$
- $\bar{x}_g^o = \bar{x}_g$
- affine transform (x^o, y^o) with new $\bar{x}_g^o$
- linearly interpolate (u, v) of f and g at new (x^o, y^o)
    - recalculate g_x from (2)
    - recalculate g_y from (3)
- update A_g with new $\bar{x}_g^o$ i.e. new (g_x, g_y)

10: Iterate steps 5-9 until $\bar{\delta} < threshold$

11: Obtain statistics
- $\bar{e} = f(u,v) - g(x,y)$
    - $x = a_0 + a_1 u + a_2 v$
    - $y = b_0 + b_1 u + b_2 v$
- $\hat{\sigma}_o^2 = \frac{\bar{e}^T e}{\mathrm{r-u}_g}$
- $C_{\bar{x}_g} = \hat{\sigma}_o^2 \left(A_g^T A_g\right)^{-1}$
- $P_{\bar{x}_g} = \frac{C_{(i,j)_{\bar{x}_g}}}{\sqrt{C_{(i,i)_{\bar{x}_g}} * C_{(j,j)_{\bar{x}_g}}}}$

At each iteration, the A_g matrix is updated. The A_g matrix highly deviates from being fixed because it is populated by $g(x, y)$ which was in fact a stochastic entity. Therefore, when iteratively updating A_g, stochastic properties of A_g cannot be ignored or else the solution will not converge or be incorrect. No other entities are updated which implies a consistent and proper estimation. This method was never implemented due to its computational requirements and iterations to the A_g matrix. Also, $g(x, y)$ is a stochastic term, and it is used to populate the A_g matrix. Once $g(x, y)$ is interpolated, its stochastic properties would deviate the estimation of the parameters. So, it was never able to work properly because the deterministic and stochastic balance in the least-squares adjustment was compromised by updates made to A_g using interpolated values.

METHOD 2 Least-Squares by Adjustments to $f(u, v)$

This method requires the least computational solicitation. At each iteration, $f(u, v)$ is affine transformed closer and closer to $g(u, v)$. Therefore g is always evaluated at (u, v), and the A_g matrix is only populated once. However, after $f(u, v)$ is transformed to some (x, y) it must be interpolated at (u, v) for the next iteration; f adjusted is $\hat{f}$. To maintain consistency, the affine transformation must always be applied to the original $f(u, v)$. So, $\hat{f}$ is **never** assigned to f during the iterations. It is important to note that this consistency is what ensures the proper estimation of $\bar{x}_g$. At each iteration, the misclosure vector is updated and $\bar{x}_g$ is adjusted. The full non-linear parametric least-squares is as follows:

1: $f(u,v) - \bar{e} = g(x,y)$

2: $f(u,v) - \bar{e} = g(x^o, y^o) + \frac{\partial g(x^o,y^o)}{\partial x} dx + \frac{\partial g(x^o,y^o)}{\partial y} dy$
- for g, $(x^o, y^o) = (u, v)$
- $dx = da_0 + uda_1 + vda_2$
- $dy = db_0 + udb_1 + vdb_2$
- g_x from (4)
- g_y from (5)

3: Affine transform (u, v) of f with $\bar{x}_g^o$
- $\begin{bmatrix} x^o \\ y^o \end{bmatrix} = \begin{bmatrix} a_1^o & a_2^o \\ b_1^o & b_2^o \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} a_0^o \\ b_0^o \end{bmatrix}$

4: Interpolate $f(x^o, y^o)$ at (u, v) and assign to → $\hat{f}(u, v)$

5: $\hat{f}(u,v) - \bar{e} = g(u,v) + g_x da_0 + g_x \bar{u} da_1 + g_x \bar{v} da_2 + g_y db_0 + g_y \bar{u} db_1 + g_y \bar{v} db_2$

6: $-\bar{e} = \bar{w} + A_g \bar{\delta}$
- $\bar{w} = g(u,v) - \hat{f}(u,v)$
- $\bar{\delta} = [da_0\ da_1\ da_2\ db_0\ db_1\ db_2]^T$

- $A_g = A_g$ from (6)

7: $\bar{e}^T\bar{e} = \left(\bar{w} + A_g\bar{\delta}\right)^T(\bar{w} + A_g\bar{\delta}) = min$

8: $\bar{\delta} = -\left(A_g^T A_g\right)^{-1}(A_g^T\bar{w})$

9: Update terms

- $\bar{x}_g = \bar{x}_g^o + \bar{\delta}$
- $\bar{x}_g^o = \bar{x}_g$

10: Iterate steps 3-9 until $\bar{\delta} < threshold$

11: Obtain statistics

- $\bar{e} = f(u,v) - g(x,y)$
  - (x,y) is calculated with the final $\bar{x}_g$
- $\hat{\sigma}_o^2 = \frac{\bar{e}^T e}{\mathrm{r-u}_g}$
- $C_{\bar{x}_g} = \hat{\sigma}_o^2\left(A_g^T A_g\right)^{-1}$
- $P_{\bar{x}_g} = \frac{C_{(i,j)_{\bar{x}_g}}}{\sqrt{C_{(i,i)_{\bar{x}_g}} * C_{(j,j)_{\bar{x}_g}}}}$

The change in pixel location for adjacent pixels is 1 and for diagonal pixels is $\sqrt{2}$. So, equations (7) and (8) for partial derivatives simplify down to

$$g_x = \frac{\partial g(x,y)}{\partial x} = \frac{\left(g(\mathrm{e}) + \frac{g(\mathrm{c}) + g(\mathrm{h})}{\sqrt{2}}\right) - \left(g(\mathrm{d}) + \frac{g(\mathrm{a}) + g(\mathrm{f})}{\sqrt{2}}\right)}{2} \quad (10)$$

$$g_y = \frac{\partial g(x,y)}{\partial y} = \frac{\left(g(\mathrm{g}) + \frac{g(\mathrm{h}) + g(\mathrm{f})}{\sqrt{2}}\right) - \left(g(\mathrm{b}) + \frac{g(\mathrm{c}) + g(\mathrm{a})}{\sqrt{2}}\right)}{2} \quad (11)$$

Further, the implemented A_g matrix simplifies to

$$A_{g_{\mathrm{rx6}}} = \begin{bmatrix} g_x(1,1) & 1g_x(1,1) & 1g_x(1,1) & g_y(1,1) & 1g_y(1,1) & 1g_y(1,1) \\ g_x(1,2) & 1g_x(1,2) & 2g_x(1,2) & g_y(1,2) & 1g_y(1,2) & 2g_y(1,2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ g_x(1,n) & 1g_x(1,n) & ng_x(1,n) & g_y(1,n) & 1g_y(1,n) & ng_y(1,n) \\ g_x(2,1) & 2g_x(2,1) & 1g_x(2,1) & g_y(2,1) & 2g_y(2,1) & 1g_y(2,1) \\ g_x(2,2) & 2g_x(2,2) & 2g_x(2,2) & g_y(2,2) & 2g_y(2,2) & 2g_y(2,2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ g_x(m,n) & mg_x(m,n) & ng_x(m,n) & g_y(m,n) & mg_y(m,n) & ng_y(m,n) \end{bmatrix}$$

When operating pixel by pixel, it can be computationally practical to use a kernel (Gonzalez, 2004). The following kernels respectively corresponding to equations (10) and (11) may be used. Note that most image filtering functions tend to truncate output, so the user should be cautious.

Kernel for g_x

$-\frac{1}{2\sqrt{2}}$	0	$\frac{1}{2\sqrt{2}}$
$-\frac{1}{2}$	0	$\frac{1}{2}$
$-\frac{1}{2\sqrt{2}}$	0	$\frac{1}{2\sqrt{2}}$

Kernel for g_y

$-\frac{1}{2\sqrt{2}}$	$-\frac{1}{2}$	$-\frac{1}{2\sqrt{2}}$
0	0	0
$\frac{1}{2\sqrt{2}}$	$\frac{1}{2}$	$\frac{1}{2\sqrt{2}}$

Although this method does not follow conventional parametric least-squares, it still works for matching images. Least-squares is only implemented to estimate $\bar{x}_g$ and this method still follows that basis. At each iteration, original $f(u,v)$ is adjusted with the, while $g(u,v)$ stays consistent. This consistency allows the stochastic properties of g to be ignored, as it is used in the design matrix. Due to the consistency of $g(u,v)$ and only the misclosures changing, $\bar{x}_g$ is estimated properly. This method maintains the deterministic and stochastic balance of least-squares adjustments.

METHOD 3 Method 2 but Using an Inverse Affine

When implementing method 2, it was found that an **inverse** affine transformation to $f(u,v)$ output the exact same results when $\bar{\delta}$ was set as $-\bar{\delta}$. So, the exact implementation of method 2 was followed, but step 3 and step 9 were changed as follows.

3: Inverse affine transform (u,v) of f with $\bar{x}_g^o$

- $\begin{bmatrix} x^o \\ y^o \end{bmatrix} = \begin{bmatrix} b_2^o & -a_2^o \\ -b_1^o & a_1^o \end{bmatrix}^{-1}\left(\begin{bmatrix} u \\ v \end{bmatrix} - \begin{bmatrix} a_0^o \\ b_0^o \end{bmatrix}\right)$

9: Update terms

- $\bar{x}_g = \bar{x}_g^o - \bar{\delta}$
- $\bar{x}_g^o = \bar{x}_g$

The reason this method worked was because $dx = -du$ and $dy = -dv$. Every time we updated $\bar{x}_g$ with $-\bar{\delta}$ we were solving for du and dv, which coincides with the inverse affine transform.

This method is displayed here to emphasise the variability of the geometric adjustment process to enhance computational benefits. It is important to show because there could be a method which has faster convergence. Also, it shows the flexibility of least-squares as it is applied to images and faces different computational numerical challenges.

F. Optimality Tests

When images are interpolated, it is the same as extending our measurements to infinity. In other words, within image boundaries, any real points can be interpolated and used as extensions from original images pixels. This has significant implications on the estimation of parameters and an optimal interpolation method had to be found. To determine an optimal interpolation method, different MATLAB functions were tested; including:

- Linear, interpolates on a linear polynomial. **Solution**
- Nearest neighbour, discontinuous.
- Natural neighbour, linear polynomial passing through input point.

Further, it was important to note that pixels points should not be extrapolated, specifically, at step 9 of Method 1 and step 4 of Method 2. Transformed (x^o, y^o) contain points that fail to fit the limits of (u, v). Therefore, the number of equations, r available for the next iteration may be lowered and any nonexistent-pixel functional model rows must be removed. A binary image was used to find available pixels and continue the adjustment.

To determine that extrapolation methods should be avoided, different extrapolation methods had to be tested; including some MATLAB functions:

- Linear, based on boundary gradients.
- Nearest neighbour.
- None, no extrapolation. **Solution**

Also, an extrapolation that brought in pixels from the full image was developed and tested. Essentially, this extrapolation added outside pixels at each iteration. The added pixels were not used to calculate δ at the current iteration but were used at the next iteration. This extrapolation method was referred to as edge leeway extrapolation because it brought in extra information from the edges. It did not work because it did not follow the principles of least-squares.

G. The Radiometric Functional Model

Once the non-linear parametric least-squares successfully estimates geometric distortions, the geometrically adjusted (x, y) is used in the radiometric functional model.

The radiometric adjustment is between coinciding pixels. The adjustment minimizes collinear brightness value differences by estimating linear radiometric anomalies. Radiometric anomalies are considered to occur from varied illuminance of the imaged object.

The math model is a radiometric shift and scale to g.

$$f(u, v) - \bar{e} = r_0 + r_1 g(x, y) \tag{12}$$

The functional model takes the linear parametric least squares form; and is derived as follows:

1: $f(u, v) - \bar{e} = r_0 + r_1 g(x, y)$
2: $-\bar{e} = \bar{w} + A_r \bar{x}_r$
- $\bar{w} = -f(u, v)$
- $\bar{x}_r = [r_0\ r_1]^T$
- A_r from (10)

3: $\bar{e}^T \bar{e} = (\bar{w} + A_r \bar{x}_r)^T (\bar{w} + A_r \bar{x}_r) = min$
4: $\bar{x}_r = -(A_r^T A_r)^{-1} (A_r^T \bar{w})$
5: Assign $r_0 + r_1 g(x, y) \rightarrow \hat{g}(x, y)$
6: Obtain Statistics
- $\bar{e} = f(u, v) - \hat{g}(x, y)$
- $\hat{\sigma}_o^2 = \frac{\bar{e}^T e}{\mathrm{r} - \mathrm{u}_r}$
- $C_{\bar{x}_r} = \hat{\sigma}_o^2 (A_r^T A_r)^{-1}$
- $P_{\bar{x}_r} = \frac{C_{(i,j)_{\bar{x}_r}}}{\sqrt{C_{(i,i)_{\bar{x}_r}} * C_{(j,j)_{\bar{x}_r}}}}$

The A matrix for the linear radiometric functional model is given by

$$A_{r_{rx2}} = \begin{bmatrix} 1 & g(x_1, y_1) \\ 1 & g(x_1, y_2) \\ \vdots & \vdots \\ 1 & g(x_1, y_n) \\ 1 & g(x_2, y_1) \\ 1 & g(x_2, y_2) \\ \vdots & \vdots \\ 1 & g(x_m, y_n) \end{bmatrix} \tag{13}$$

Results

A. Accuracy of Single Target Sub-Images

Method 2 was tested on 14 single target sub-images. The set of targets had varying geometric and radiometric quality. Target 5 and generated target templates were set as f, then matched to sub-images separately.

Convergence plots give valuable information on the correlation of parameters and overparameterization. Oscillating convergence plots as in figure 8, high correlations and visually incorrect estimates of target centers suggested overparameterization from six affine transformation parameters for elliptical targets.

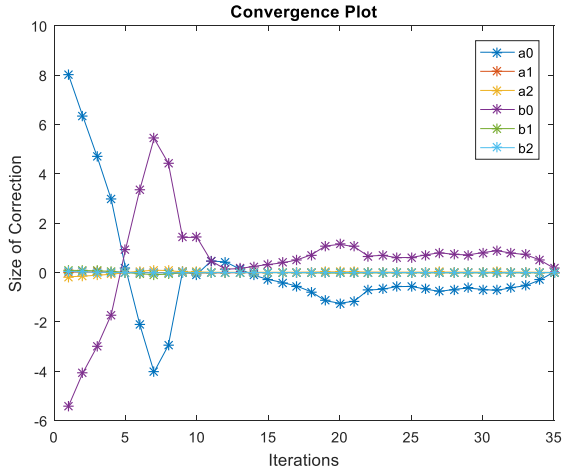


Figure 8. The convergence plot has six geometric parameters and ossilates around zero.

An ellipse that is rotated and scaled can also be sheered to provide the same orientation. So, five parameters account for all the possible orientations for ellipses. Either a_2 or b_1 was not used. And, five geometric parameters provided minimal oscillations in convergence plots and lower correlations, and visually more accurate target center estimations. Therefore, their pixel rms values could be trusted. Further, it was found that the Template only fit the match better for targets 217 and 195. This was because, gray value variations of the template were more like that of targets 217 and 195. Refer to Appendix I for the full set of combination of parameter results.

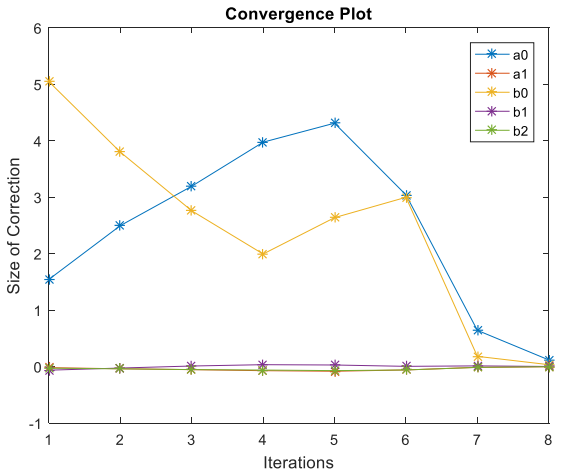


Figure 9. The bottom plot has five geometric parameters and has a smooth convergence to zero.

For some convergence plots the $\bar{\delta}$ adjustment values were smaller at the initial iterations and then larger in the middle. This is because the program is getting closer and closer to the solution until a concrete estimation of $\bar{\delta}$ can be made. Supported by the fact that the initial approximations are always [010001], which would require higher number of iterations for some adjustments.

The template was not developed enough to model certain matches better than Target 5. It would be difficult to make a radiometric template that matches every target the best. Simply, targets match each other better when they are taken from similar target locations because they have the most radiometric similarity.

Table 1. Pixel rms precision of match for Target 5 and Template against 14 varied sub-images.

Target	RMS Target 5		RMS Template		Target 5 matched better?
	x var (pix)	y var (pix)	x var (pix)	y var (pix)	
5	0.000	0.000	0.002	0.006	✓
53	0.002	0.004	0.006	0.012	✓
191	0.002	0.007	0.003	0.011	✓
4	0.000	0.002	0.001	0.005	✓
167	0.002	0.008	0.004	0.017	✓
217	0.025	0.072	0.007	0.020	×
15	0.000	0.001	0.002	0.006	✓
6	0.001	0.001	0.002	0.005	✓
195	0.014	0.047	0.004	0.014	×
171	0.002	0.007	0.003	0.008	✓
174	0.001	0.004	0.003	0.008	✓
8	0.001	0.004	0.002	0.008	✓
59	0.001	0.006	0.005	0.021	✓
68	0.001	0.005	0.004	0.015	✓

Target 5 was matched to a translated version of itself. This meant that the radiometry of g and f were completely similar, so the design matrix completely modelled the situation. Expectedly, the match provided pixel rms values of 0.000. From Appendix I, targets 4 and 6 were extracted from a region proximal to target 5, which also meant high radiometric similarity, therefore low rms. Vise versa, targets such as 59 and 68 were distal and matched unwell to both Target 5 and the Template. However, when target 59 was matched to more proximal targets, i.e. target 68, they matched well due to their radiometric similarity. Therefore, targets that were proximal to each other had ideal convergence plots, lower parameter correlations and lower rms values compared to matches between distal targets.

Some image matches had a bias in their error probability distribution histogram (PDH) which was due to f and g **not** being similar in radiometry. This can be seen in figure 10. The template was only made geometrically similar, so it could be developed in radiometric proficiency to obtain a better PDH and more reliable accuracy values. This bias is further discussed in the 2nd recommendation.

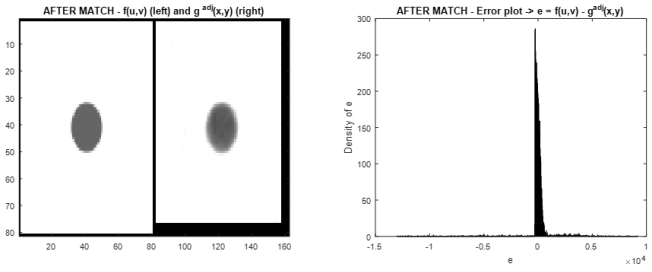


Figure 10. Error probabilty distribution histogram with a bias, due to dissimilarities in brightness value distribution. Template is $f(u, v)$.

High radiometric similarity between matching targets had error PDHs that resembled normal distributions like in figure 11.

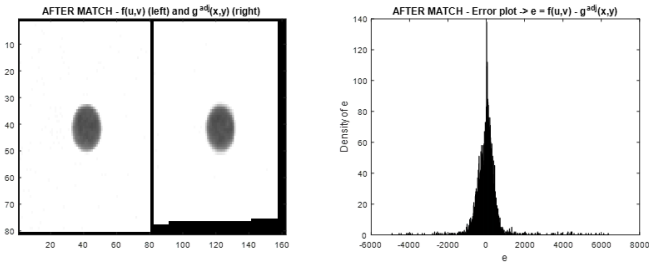


Figure 11. Error probabilty distribution histogram with no bias, due to similarities in brightness value distribution. Target 5 is $f(u, v)$.

Method 2 was computationally simple and estimated the geometric distortion parameters properly. The least-squares method is a method which should impose the sum of residuals to be zero. Even if the template and slave are different, the sum of residuals should be zero given that a correct least squares process was followed. The final error was zero for Method 2. The computational requirements and processing time of least-squares matching may be a problem in terms of feasibility. Interpolation and transformation especially take long. However, the number of iterations could drastically be lowered with a good point of expansion.

To conclude, if one can mathematically model the 2D movement of an image, LSM can be used to track any pixel with a precision of sub 0.1 pixels. Method 2 matched the images more reliably. Least-squares matching could be feasible in estimating target centers and transforming one image to another without explicitly selecting any tie points.

Limitations and Future Work

1. The matched images can only be described by single plane transformations. If we have two separate planes projected on to a single image plane, how can we distinguish the two mathematically? The single image plane will be a function of two separate object planes. Furthermore, the image is going to transform between each sequential image based on how the two separate planes are moving. Therefore, the two separate object planes, depending on their coordinates, will overlap. This can be seen in the full target image in Appendix I. So, how can we model these overlapping of planes, i.e. the overlapping of the sides of the cubic object ingrained with circular targets? In physics, waves overlap and make different amplitudes. The separate planes could be modelled like waves; however, their brightness values will not be amplified when they overlap but one brightness value will make the other disappear due to opacity. Basically, there will be two separate signals that converge and diverge as the cubic object rotates. If signal processing is used and one signal from the corresponding plane is extracted, then the separated plane could be replotted! This would be initial work on the cubic calibration images. Then the same least-squares matching method could be applied.
2. The radiometric anomaly parameters can only apply linear radiometric corrections to the image. If the radiometric anomalies could be parametrized to a point where adjustments are specific to a pixel, we would be able to get rid of the bias when matching the template. As seen before, when matching the

geometrically similar template and slave there was a bias due to brightness values of the template being constant everywhere but their edges. To get rid of this bias, the template would need to be modelled by non-linear radiometric parameters. However, this may be undesirable because non-linear radiometric anomalies seldom need to be modelled.

3. Least-squares matching may be implemented as a window function that moves over the image.
4. Given a full cubic calibration image, is the affine movement between the center pixels the same as the pixels distal from the center of the image? This may be tested by obtaining the parameters from running sequential LSM on multiple targets taken from the center of the image, and multiple targets taken away from the center. The next step would be to compare the two separate sets of parameters.
5. If appropriate to the DF system, use targets that are not circular. Targets that will not be overparametrized by the full affine parameter set will require less iterations, and a more reliable adjustment.
6. The computational processing time could be drastically lowered for interpolation, image transformations, and matrix inverses by utilizing parallel programming. All three computational tasks are parallelizable and have developed methodologies for implementations. Furthermore, MATLAB for GPU computing is already possible. This could potentially increase the feasibility of LSM by speeding up computation time.

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